

ARYAN ZAVERI

Quantitative Developer | Cross-Asset Research | End-to-End Trading Infrastructure

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TARGET PROFILE

Quantitative Developer building systems that treat markets as connected decision surfaces: global sectors, macro variables, Indian market structure, options/futures behavior, crypto risk appetite and execution quality feed the research and implementation loop. Strongest work combines research judgment with implementation: quantify relationships, test them, simulate them, replay them, and wire the surviving logic into risk and execution infrastructure.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Shell, JavaScript

Trading Systems: market-data ingestion, live state, feature engineering, signal generation, strategy research, backtesting, walk-forward validation, replay, paper-fill, pre-trade risk, order intents, execution journals, post-trade analytics

Research Surface: Indian/global sector relationships, crude oil and macro themes, US/China/Japan/Korea market links, sector rotation, breadth/regime detection, pullback-entry and strength-exit behavior, cross-asset correlation, options/futures relationships

Data/Infra: PostgreSQL, Redis, SQLite, JSONL, CSV/TSV archives, Linux, AWS, Git, Docker, WebSockets, REST APIs

Metrics: Sharpe, Sortino, profit factor, win rate, drawdown, expectancy, fees, fill rate, margin, deployed capital, capital required, side equity

PROFESSIONAL EXPERIENCE

GreyOak Capital - Quantitative Developer | Mar 2026 - Present

- Build quantitative research systems, market-data services, market-intelligence infrastructure and systematic-trading tooling across Python/Linux/PostgreSQL/Redis workflows.
- Own research-to-runtime handoffs across historical archives, live state, strategy/risk contracts, signal outputs, runtime logs, validation reports and production-readiness checks.
- Document service contracts, WebSocket data flows, trading contracts, validation outputs and operational handoffs for production-facing market systems.

KIFS Trade Capital - Quantitative / Algorithmic Trading Developer | 2024 - Mar 2026

- Developed algorithmic-trading tools, broker/API workflows, market-data processing scripts, research utilities and trading analytics infrastructure.
- Built intraday/options analytics using price action, indicators, option-chain features, IV/OI/Greeks-style inputs, signal filters and post-market validation.
- Implemented reusable net-PnL evaluation modules covering win rate, profit factor, drawdown, Sortino, expectancy, charges, trade count and trade-quality gates.

Greeksoft Technologies - Algorithmic Trading Intern

- Worked on algorithmic-trading workflows with exposure to broker APIs, market data, automated trading systems and exchange-facing operations.

VAJRA - PERSONAL MULTI-ASSET QUANT RESEARCH & EXECUTION PLATFORM

Independent personal platform; not a company. Proprietary strategy rules, credentials and raw order identifiers are intentionally omitted.

Research thesis and market framework

- Built Vajra around a global-to-local research framework: evaluate how global indices, sector leaders, US/China/Japan/Korea market moves, crude oil, FX, global macro and Indian macro conditions transmit into Indian sectors, stocks, futures/options behavior and crypto risk appetite.
- Focused on quantifying relationships rather than relying on discretionary narratives: correlations, lead/lag behavior, breadth, sector themes, volatility regimes, option-chain structure, cross-asset stress, capital usage, execution quality and replay/live feedback.
- Designed multi-horizon decision models that separate live execution context, intraday signal selection, daily/weekly regime and theme selection, and monthly robustness/capital review.
- Modeled action choices including continuation, reversal, pullback-entry, strength-exit, mean reversion, volatility expansion/compression, options behavior, sector rotation and risk-off/risk-on state changes.

Research-to-execution architecture

- Implemented the full loop from idea to runtime evidence: market-data archive -> feature generation -> cross-market relationship analysis -> strategy hypothesis -> walk-forward/backtest -> risk contract -> signal file -> risk decision -> order intent -> execution journal -> replay/PnL -> model refinement.
- Used Python for research/orchestration/replay, C++ for live matrix/feature/scoring paths, Redis for live state, PostgreSQL/SQLite for candles and archives, JSONL for lifecycle journals and Linux/AWS for runtime operations.
- Maintained reproducible evidence artifacts across Markdown reports, JSON summaries, CSV metrics, model artifacts, signal files, risk logs, order-intent logs, exchange journals and private socket events.

Strategy and feature surface

- Researched families across momentum, reversal, mean reversion, volatility/options, breadth, regime detection, sector rotation, cross-asset relationships, portfolio/risk overlays and execution-aware trading.
- Built equity/futures relationship strategies using market/sector anchors, side-specific rules, state-risk overlays and rolling train/test separation.
- Built options research pipelines for index/equity call-put workflows using OI, IV, Greeks-style features, expiry, moneyness, price movement and intraday option behavior.

Validation and simulation evidence

- Built rolling walk-forward engine using completed-prior-data separation and 730-day train lookback. One major 2020-01-01 to 2026-06-11 research report covered 70,013 simulated trades, 59.08% win rate, 1.386 profit factor, 58/67 positive months, explicit simulated PnL, adverse MTM drawdown and Sharpe tracking.
- Built rule/fold/month/capital summaries to evaluate robustness beyond headline PnL, including worst month, positive/negative

months, required capital, deployed margin, MTM drawdown, Sharpe and Sortino.

- Built fee-aware minute replay using live-current signals, historical minute data, risk contracts, leverage/margin constraints and exit attribution. Sampled replay accepted 2,618 trades, closed 2,613, covered 468 symbols and tracked fees, drawdown, side equity and open positions.
- Built paper-fill live-window simulation; sampled run accepted/closed 318 trades and compared profile-based capital/risk behavior with -9.57% max drawdown in the paper-fill summary.
- Built execution-strategy derivation experiments across 37 variants comparing fill assumptions, fees, fill rate, closed trades, unresolved positions, side outcomes, Sharpe and drawdown.

Live/replay/execution evidence

- Built crypto continuous-v2 infrastructure connecting 15m archives, live/current signal files, weekly model artifacts, strategy contracts, risk contracts, order intents, smart-execution journals, exchange journals and private socket logs.
- Generated structured lifecycle logs for signal generation, candidate selection, risk decisions, order intents, exchange submission paths, exit journals, smart-execution decisions and private order socket events.
- Sampled live-system evidence included 129k+ signal records, 609 risk decisions and 609 order intents in a day.
- Risk-decision schema covered side capital, projected exposure, leverage availability, score thresholds, state multipliers, rejection reasons, trading flags and risk-contract identifiers.
- Order-intent schema covered direction, margin, notional, leverage, entry price, side capital, signal IDs, risk-decision IDs, execution flags and venue/account metadata.
- Smart-execution journals tracked preflight readiness, duplicate-intent checks, exchange gates, slippage constraints, laddering fields and exchange-send status.

Risk and portfolio controls

- Built pre-trade and portfolio controls around side capital, side-share limits, deployable capital, leverage availability, buffer capital, open positions, one-position-per-symbol, drawdown states, caution/defense/pause profiles and stop/trailing/time exits.
- Built reports tracking peak deployed margin, adverse MTM drawdown, required capital, worst month, positive/negative months and side-specific performance.

EDUCATION

Graduate, 2024.

EVIDENCE BOUNDARY

Research, simulation, replay, paper-fill and live-system activity are documented separately. Audited live trading profitability is not claimed. Proprietary strategy rules, credentials and raw order identifiers are intentionally omitted.